Revisiting acoustic inference

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Outline

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3) Pietarinen’s acoustic logic
4) Champagne on acoustic logic
5) A revised scheme
6) Strawson on acoustic experience
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Acoustic logic

Traditionally, logic has made use of linguistic media (either in written text, or thrugh speech).

The advancement of modern logic came with an examination of different media for expressing logical structure and arguments.

- See, for example, Peirce's diagrammatic logic and Frege's *Begriffsschrift*.

There is a growing interest in visual logical diagrams. But what about other media?

Here, we will explore the possibilities for **acoustic logics**, logics that use sound as the medium for expressing logical structure.

There is a certain tradition in informal logic that studies the logical properties of sounds: from the occurrence of a loud bang, we infer that a bomb exploded nearby.



There is also a purely *formal* study of sound as a logical medium. I will call this **pure acoustic logic**, and it will be our focus here.

Leading questions

1) Can a purely acoustic medium represent logical structure?
2) Can a purely acoustic medium represent an argument?
3) What is necessary for an agent to carry out inference through such representations?

The first half of this talk will be an approach to the first two questions, and the second will attack the third.

Looking at Peircean diagrammatic logic

The idea of a pure acoustic logic has been explored preliminarily by Peircean scholars.

In his 1892 paper, Peirce analogizes the carrying out of an argument with certain manipulations in a diagram. He says:

> […] this diagram must be so contrived that it is easy to examine it and find out whether the course of the river is in truth in every case such as is here proposed to be inferred. **Such a diagram has got to be either auditory or visual**, the parts being separated in the one case in time, in the other in space. […]
>
> —

Peirce talks here of ‘auditory diagrams’. I suspect he is talking about speech.

Pietarinen (2010) takes this as a hint to explore an acoustic analogue to Peirce’s visual diagrammatic logic.

Peircean diagrams, the basics

- Primitives: the *sheet of assertion*, over which it is possible to make *inscriptions* and *cuts*.
- The empty sheet represents logical truth ($\top$).
- Inscriptions represent assertibles (propositions).
- Conjunction is joint inscription.
- Negation is inscription within a *cut*.

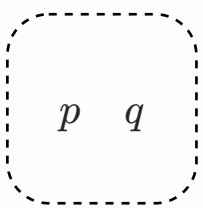


Figure 1: $p \wedge q$

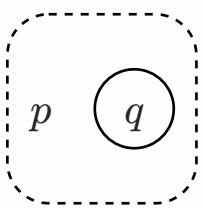


Figure 2: $p \wedge \neg q$

- The disjunction of p and q is the negation of the conjunction of $\neg p$ and $\neg q$.
- Proving a formula as tautological is the same as reducing it to the empty sheet of assertion.
- The propositional system (Peirce calls it the system of alpha graphs) is equivalent to classical propositional logic.

Pietarinen's acoustic logic

- The idea is to develop acoustic analogues to the sheet of assertion, inscriptions and cuts.
- The sheet of assertion is temporally extended, a sonic landscape of indistinct features against which inscriptions can be made.
- Inscriptions are tones with a delimited frequency.
  - Natural sounds are really stacks of waves at a multitude of frequencies (harmonics). The logic abstracts away from this.
- Over white noise, tones are distinguished by loudness, pitch and timbre.
- Multiple sounds can occur at the same time–this will be a *chord*, and will correspond to a conjunction.
- Negation needs a different device: Pietarinen proposes that negation is an inscription in the acoustic sheet of assertion that signifies that some tone is not inscribed in the sheet of assertion. The tone must not be distinctly heard: there is a silence at the frequency of the tone.
  - Because mutes silence frequencies, all tones with the same frequency must be equivalent (they have the same negation).
  - Mutes can be muted: in that case, the original tone should be heard (classically).
  - The negation of tautology is absolute silence.

A graphical representation for Pietarinen's acoustic logic

- To make things clearer, we can make use of visual diagrams that represent these structures.
- This requires care: visual and acoustic diagrams are not perfectly analogous, and must be interpreted.
- The sheet of assertion is a gray surface over which inscriptions can be made. Its y axis represents frequency, and its x axis represents time.
- Tones are dark horizontal lines at a given frequency.
- Mutes are white horizontal lines at a given frequency.

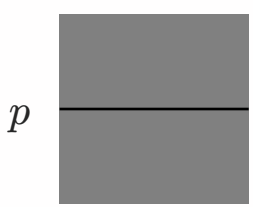


Figure 3: p



Figure 4: $\neg p$

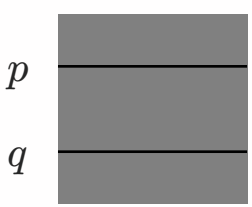


Figure 5: $p \wedge q$

Disjunction

- The negation of a chord is equiovalent to the disjunction of the negation of each component ($\neg(p \wedge q) \equiv \neg p \vee \neg q$, as De Morgan's law).
- Pietarinen says:

> A disjunction of two tones is a mute of a chord consisting of two muted tones, and an implication between two tones is a mute of a chord consisting of a 'heard' tone (the antecedent) and a muted tone (the consequent).
>
> — p. 78

  As we will see, this will bring trouble.
- In the meanwhile, think about this question: where are the *cuts*?

Inference

- We capture inference as chains of combinations of chords and mutes (*phrases*).
- A *bar* is an extension of time in which a phrase can be heard of muted.
- A heard bar is not muted or muted an even number of times.
- A silent bar is muted an odd number of times.
- A bar *follows* from another if it is the result of some of these operations:
  1) Any sound can be doubly muted, and any doubly muted sound is a sound.
  2) Any sound can be deleted from a heard bar.
  3) Any sound can be added to a silent bar.
  4) Any sound can be copied to/deleted from a subsequent bar.
- These are analogous to Peirce's rules for alpha graphs.

Champagne on acoustic logic

Champagne (2015) offers a revision of Pietarinen's logic.

- Here, the sheet of assertion is an acoustic plenum, a sonic landscape where every frequency can be heard.
- Tones substract from the plenum.
- A mute of a tone substract the tone from the plenum.
- The diagrams Champagne uses do not distinguish between tones and the background.

Figure 6: a tone

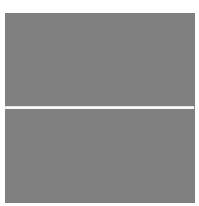


Figure 7: the mute of a tone

Sequential conjunction and inference

- Champagne also introduces a notion of sequential conjunction.

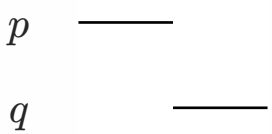


Figure 8: p, q

- This will be useful for the display of arguments:

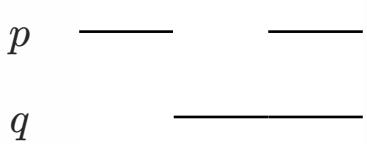


Figure 9: $p, q \vdash p \wedge q$

- The conclusion here is supposed to follow from the premises through the application of a rule like Pietarinen's rule 4.

Implication

- Champagne uses sequential conjunction to reconstruct implication.
- Classically, $p \to q \equiv \neg(p \wedge \neg q)$, so it is the mute of the sequential conjunction of p and $\neg q$.
- The mute of a sequence of assertions is the the sequence of the negations of those assertions.



Figure 10: $p, \neg q$



Figure 11: $\neg(p, \neg q)$ (?)

Problems!

- Take *modus ponens*:



Figure 12: $p \to q, p$

- Here, we have tones that are silent in one bar and muted in another. These are sequential contradictions!
- The problem also applies to double negations: if each step of the production of a double mute is inscribed in the sheet of assertion, we will also get sequential contradictions.

The mistake

- Treating implication in this way is a mistake:
  - The problem is that $p \to q$ cannot be equivalent to the sequential conjuntion of $\neg p$ and q.
  - $\neg p \wedge q$ represents only one of several conditions in which $p \to q$ is true; we lose the connection between the conditional ad disjunction.
- Reasoning like Champagne, $p \vee q$ would be equivalent to $\neg(\neg p \wedge q)$:



Figure 13: $\neg p, \neg q$



Figure 14: $\Rightarrow\!\!\!|(\neg p, \neg q)$ (?)

- But this is not classically correct!
- Contrary to what the definability of $\vee$ in terms of $\neg$ and $\wedge$ would suggest, here having the resources for expressing negations and conjunctions is not sufficient to capture ways to express disjunctions and implications.

A revised scheme

- These issues can, I will argue, be aleviated revising the scheme.
- One fundamental change is to treat tones as schematic variables, not as concrete atomic propositions. The formal features of sound sequences should be transposable across the whole range of frequencies.
- We go back to Pietarinen's approach to the sheet of assertion and basic inscriptions.
- I will approach the issue visually first, to then offer an acoustic interpretation.

Disjunction, revisited

- We will represent disjunctions by *stacks* of phrases, separated by dotted lines or cuts.
- The idea is analogous to that of tracks of audio mixed together.
- While in chords sounds are asserted together ($p \wedge q \vdash p$ and $p \wedge q \vdash q$), in stacks no component of the stack is asserted ($p \vee q \nvdash p$ and $p \vee q \nvdash q$).

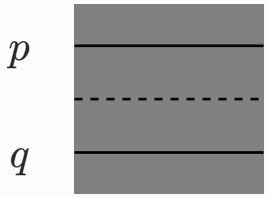


Figure 15: $p \vee q$

- We get De Morgan's laws quite naturally:



Figure 16: Some De Morgan's laws

- Disjunction is naturally commutative.

Conditionals revisited

- The conditional becomes straightforward:

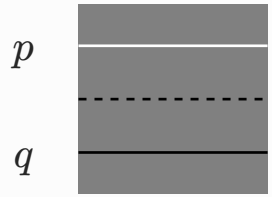


Figure 17: $\neg(p \wedge \neg q) \equiv \neg p \vee q \equiv p \rightarrow q$

- *Modus ponens* avoids the issue Champagne raised:

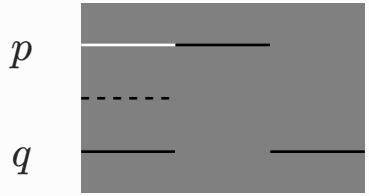


Figure 18: $p \rightarrow q, p \vdash q$

There is still a problem, but we will return to that later.

In the first phrase, $\neg p$ is not asserted, so there is no sequential contradiction.

Tautologies

- We should be able to reconstruct $((p \to q) \wedge p) \to q$ as a tautology.
- To do this, we need to add a new device to represent the degree of subformulas (otherwise the constructions will be ambiguous, given the commutativity of disjunction).
- The degree of the subformula can be represented as a difference in the intensity of the background of the sheet of assertion.
- A formula of degree n will have intensity $\frac{1}{n}$; so degree 1 formulas will have an intensity $\frac{1}{1}$, formulas of degree 2 will have an intensity $\frac{1}{2}$, and so on.
- We get:

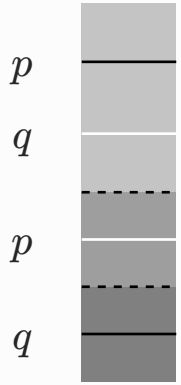


Figure 19: $((p \to q) \wedge p) \to q$

Assuming density (between every frequency there is another frequency), this device allows the construction of formulas of arbitrary complexity.

From phrases to sequences and back

- What we want is some kind of analogue to the deduction theorem: in this way, we can prove tautologies by validating sequences, and validating sequences proving tautologies.
- Our previous discussion suggests:

  Deduction A sequence $\Pi_1, ...\Pi_n, \psi$ is such that $\Pi_1, ..., \Pi_n \vdash \psi$ if and only if the phrase $\neg\Pi_1 : ... : \neg\Pi_n : \psi$ is a tautology.

- A phrase diagram represents a tautology either when it can be constructed from the tautology-sound following a series of rules like Pietarinen's, or when it can be reduced to the tautology-sound.
- Alternatively, the principle we just stated says that we can say that a phrase diagram is a tautology if it can be converted from a phrase-sequence-diagram such that its last phrase is a valid inscription given the previous phrases.

The acoustic interpretation

- OK, so the prroblems with Pietarinen and Champagne acoustic logics can be solved.
- The solution is visual. Is there an acoustic interpretation of this that works?
- Disjunction poses a problem: bare unison inscription cannot represent disjunction. If the parts of disjunction coincide temporally, as their stacking suggests, this is a problem.
- A partial solution: tones in each part of the stack should have a different ‘character’ (timbre or envelope), while for tones it should remain the same.
  - This deals with the distinction between $p \wedge q$ and $p \vee q$, but not for $p \wedge \neg q$ and $p \vee \neg q$ (which are acoustically indistinguishable as silences do not have a character).

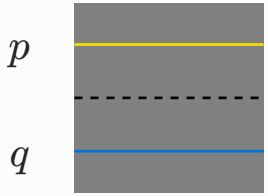
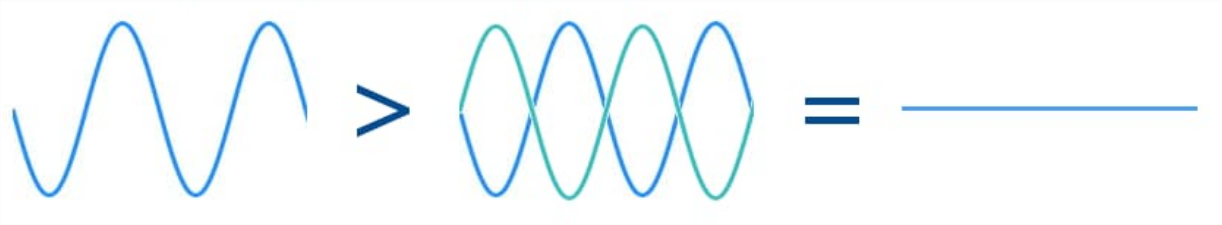


Figure 20: $p \vee q$

Revisiting mutes

- Mutes should not be understood as silences, but as other kinds of systematically correlated sounds. For example, the mute of a phrase could be an identical phrase with opposite phase. The result of adding up two signals of opposite phase cancel each other through destructive interference.



- So p and $\neg p$ lead to $\bot$ (silence).
- This allows us to distinguish between $p \wedge \neg q$ and $p \vee \neg q$ ($\neg q$ in the first and the second have different timbre).

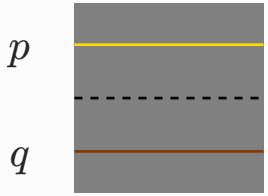


Figure 21: $p \vee \neg q$

Going deeper

- So we have an initial case for saying that something like pure acoustic logic is viable.
- Could this be a way for agents to reason? Would agents who only had access to auditory experience have the capacity to think logically though means like these?
- Champagne notices that someone could argue:
  1) Spatial concepts are fundamental (Lakoff & Johnson 1995)
  2) The experience of sound alone does not provide the resources for spatial concepts (Strawson 1959)
  3) Auditory experience alone cannot provide the resources required for pure acoustic logic.
- How autonomous, from a conceptual point of view, is pure acoustic logic?

Strawson on auditory experience

> The fact is that where sense-experience is not only auditory in character, but also at least tactual and kinaesthetic as well—or, as it is in most cases, tactual and kinaesthetic and visual as well—we can then sometimes assign spatial predicates on the strength of hearing alone. But from this fact it does not follow that where experience is supposed to be exclusively auditory in character, there would be any place for spatial concepts at all. I think it is obvious that there would be no such place. The only objects of sense-experience would be sounds. Sounds of course have temporal relations to each other, and may vary in character in certain ways: in loudness, pitch and timbre. But they have no intrinsic spatial characteristics: such expressions as 'to the left of', 'spatially above', 'nearer', 'farther' have no intrinsically auditory significance.
>
> — Strawson (1959), p. 65

- This yields a non-spatial view of auditory experience.
- The question here is whether this non-spatial conception can still account for the possibility of purely auditorily-based logical thought.
- Think of the question: how would being whose experience was purely auditory be able to think logically?

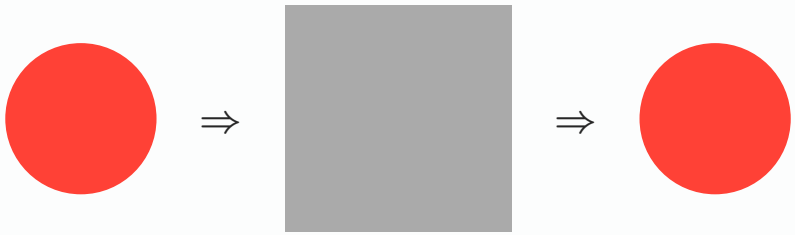
Reidentifiability as a requirement for logical thought

- A minimal requirement for logical thought to be possible is that the relevant structures can be held constant through the whole process of thinking an argument of evaluating the properties of some structure.
  - In the inference from $p \wedge q$ adn p to q, p and q must mean the same thing each time, and this requires the ability to reidentify the relevant representations.
- In the acoustic case, sonic inscriptions must remain in some sense reidentifiable through time, and it is plausible that the auditory medium has different resources to enable this than the visual medium.

Are logical inscriptions particulars or types?

- Consider the following visual experience:

  There is a gray box in front of you. From the left, a red ball slowly rolls behind the box. After a while, a red ball slowly rolls from behind the box, at roughly the same speed as the first one. Are they the same particular ball? They may or may not be (the ball could have been replaced behind the box, or the first ball could have pushed a second when behind the box, which is what we see roll out); nonetheless, we can identify them as belonging to the same kind (that of red balls).



- In the acoustic case, we might ask whether occurring tones should be reidentified as particulars that repeatedly appear or as repeating types.
- Strawson (1959) thinks that type reidentification is perfectly OK, so the problem is solved if that is all that acoustic logic requires (and if Strawson's is correct).

Would auditory experience yield a concept of something universal?

- It is not clear to me that we can make sense of the notion of something universal purely from auditory experience.
- We can, I think, make sense of repeatable particulars auditorily.
- How could an agent whose experience was purely auditory come up with a *notion* of a universal that is different from that of a repeatably occurrent particular?
- It then seems that in the auditory case we have to revert to considering particular reidentification independently of universal reidentification.

Strawson on auditory particular reidentification

- Strawson does provide something like what we want.
- Imagine a *master-sound* that is heard continuously at a constant intensity, with a varying pitch. Besides it, other sound sequences can be heard. They recur and can be recognized.

> [...] suppose a particular instance of one of the unitary sound-sequences I mentioned is being heard. A gradual change in the pitch of the master-sound is accompanied by a gradual decrease, or a gradual increase followed by a gradual decrease, in the loudness of the unitary sound-sequence in question until it is no longer heard. If the gradual change in pitch of the master-sound continues in the same direction, a different unitary sound-sequence is head with gradually increasing loudness. [...]
>
> — p. 75

> The pitch of the master-sound at any moment would determine the auditory analogue of position in the sound-world at that moment. The sound-world is then conceived of as containing many particulars, unheard at any moment, but audible at other positions than the one occupied at that moment.
>
> — p. 76

Master-sounds in pure acoustic logic

- In Peircean acoustic logic, the role of the master-sound in the Strawsonian scheme can be played by tautology sounds, which serve as the background for every inscription.
- The possibility of changes in the master sound is necessary to make particular reidentification possible in the scheme (cf. Strawson 1959, p. 79–80).
- However, it seems as if changes in the master sound are precluded by the Peircean idea that the tautology sound is something indistinct.
  - If we accept this, our options as long as we want to allow reidentification are:
    - to concede that the relevant kind of reidentification is universal reidentification,
    - abandon the idea that the resources for acoustic logic are autonomous.
- But! When we examined the possibility of capturing hierarchical logical structure, we already moved in the direction of allowing changes to the auditory sheet of assertion.

What acoustic inscriptions may mean and the taking condition

- These issues affect not only our capacity to reidentify logical structure, but also our capacity to distinguish the *role* of different structures in arguments.
- Boghossian (2014) proposes that inference requires that the subject who executes it is able to distinguish between premises and conclusions.

  Taking condition the thinker should take his premises to suppport the conclusion and draw the conclusion because of that fact.

- We only need to concern ourselves with something even more basic:

  Distinguishability The role of support and conclusion must be distinguisable in the structures that serve as carriers of inference.

Illation and distinguishability

- Consider how, in an acoustic setting, we would distinguish between $p \to q, p \vdash q$ and $p \to q, p, q$.
  - The first is a valid inference, while the second is merely a sequence of assertions.
- We can capture the different if we allow sequential shifts in the background of the sheet of assertion. We can distinguish premises from conclusions by varying the backgrounf of condlusions.
- Thus, $p \to q, p \vdash q$ would become:

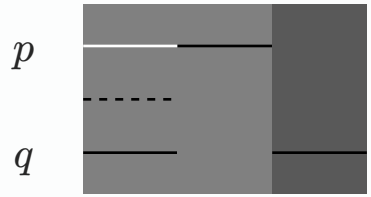


Figure 23: $p \to q, p \vdash q$ (revised)

- We can only know what plays what role once the sequence has ended.

A problem with $\perp$

- Suppose we tried to diagram a complex argument with steps that involve something like negation introduction or elimination.
- We would, then, have sequences with $\perp$ as intermediate steps.
- In visual natural deduction systems we have ways to represent hierarchy so that contrradictions do not escape to the top level of proofs, but in this system $\perp$ is always represented by silence, and all silences are perceptually the same.
- To distinguish the role of $\perp$, we need to look at the context.

Another idea from Peirce: growth

- Champagne & Pietarinen (2019), in an article that argues that static pictures cannot represent arguments while moving ones can, appeal to an idea from Rose (2011): *growth*.
- Growth is a form of determination where what grows from something else evolves from it.
- Applied to illation: a conclusion is something that has grown from something else, while a premise is something from which something else grows.
- The hierarchical structure is grasped insofar as the dependencies between its sequential parts are understood.
- The growth of auditory signs can only happen as a process that extends over time. While auditory landscapes can offer auditory structure, only auditory sequences can express arguments.
- What we look for in the context of arguments in order to capture its sequential hierarchy between premises and conclusion are the relevant growth-depencence links.

Disjunction and modality

- One last thing: the conceptual resources for dealing with disjunction are also plausible precursors for those required for modalities.
- The ‘cuts’ that acoustic disjunction seems to require also offer a model for dealing with possibilities.
  - This is similar to the status of modal cuts in Peircean logics (for example, what happens with gamma-graphs).
- Rather than presenting inscriptions as assertible *here*, modal cuts present them as *assertible somewhere*.
- What is acoustically possible is that can be heard either *here* or *there*.
- This makes sense if we think of modality as something that involves ‘spaces’ in some sense.
- The sheet of assertion becomes a three- or multi-dimensional space for inscriptions.

Conclusion

- Pure acoustic logic presents a unique and fascinating set of challenges.
- It can provide a model for subclassical logics where expressive resources are constrained.
  - Maybe not too interesting on its own: after all, we can examine subclassical logics like these by simply tweaking the proof systems of the logics we are interested.
  - But what acoustic logics provide is an *interpretation* for those subclassical logics.
- It offers a model for examining the conditions that make logical thought possible across different media.
  - Here, I have used them as an exploratory model.

¡Gracias!



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